Muta-Gene® Phagemid *In Vitro* Mutagenesis Version 2

Instruction Manual

Catalog Number 170-3581

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Foreword

This manual contains background information and detailed protocols for performing *in vitro* mutagenesis with the Muta-Gene phagemid *in vitro* mutagenesis kit. The mutagenesis technique described is intrinsically very easy to use. It requires only one enzymatic step; the two bacterial strains are healthy and easy to grow; and the efficiency of mutagenesis is usually sufficient that mutants can be identified by DNA sequence analysis rather than a phenotypic or hybridization screen.

Bio-Rad's kit provides reagents of the highest quality. All components have been rigorously tested in *in vitro* mutagenesis experiments. If you have any questions regarding the use of these products, please contact your local Bio-Rad representative or Bio-Rad's Technical Services at 1-800-4BIORAD.

These products are for research use only and not for use in humans or for diagnostic procedures.

Safety precautions should be used when handling hazardous materials (radioactive nucleotides, acrylamide, ethidium bromide, phenol, chloroform, ether, etc.).

Users of this kit should comply with NIH or other relevant guidelines for recombinant DNA work.

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Section 1
Introduction

Oligonucleotide directed *in vitro* mutagenesis is a widely used procedure for the study of the structure and function of DNA and its encoded protein. Many techniques are available for performing *in vitro* mutagenesis (reviewed in reference 1). A typical strategy is to clone the segment of DNA to be mutated into a vector whose DNA exists in both single- and double-stranded forms. An oligonucleotide complementary to the region to be altered, except for a limited internal mismatch, is hybridized to a single-stranded copy of the DNA. A complementary strand is then synthesized by T7 DNA polymerase using the oligonucleotide as a primer. Ligase is used to seal the new strand to the 5' end of the oligonucleotide. The double-stranded DNA, completely homologous except for the intended mutation, is then transformed into *E. coli*, resulting in two classes of progeny—the parental and those carrying the oligonucleotide-directed mutation. Since there are both parental and mutant progeny, no more than half of the progeny will be mutant. In practice, a much lower fraction is usually obtained.

1.1 Principle of the Method

The Muta-Gene phagemid *in vitro* mutagenesis kit is based on a method described by Kunkel[2,3] which provides a very strong selection against the non-mutagenized strand of a double-stranded DNA. When DNA is synthesized in a *dut ung* double mutant bacterium, the nascent DNA carries a number of uracils substituted for thymine as a result of the *dut* mutation, which inactivates the enzyme dUTPase and results in high intracellular levels of dUTP. The *ung* mutation inactivates uracil N-glycosylase, which allows the incorporated uracil to remain in the DNA. This uracil-containing strand is then used as the template for the *in vitro* synthesis of a complementary strand primed by an oligonucleotide containing the desired mutation. When the resulting double-stranded DNA is transformed into a cell with a proficient uracil N-glycosylase, the uracil-containing strand is inactivated with high efficiency, leaving the non-uracil-containing survivor to replicate. Typical mutagenesis frequencies obtained with the Muta-Gene phagemid kit are greater than 50%, a rate high enough to allow direct identification of mutants by sequence analysis.

1.2 Phagemid and Helper Phage

In order to perform *in vitro* mutagenesis by this method, a vector that can exist in both double- and single-stranded forms is required. Previously, the vectors most often used in *in vitro* mutagenesis have been derivatives of the *E. coli* bacteriophage M13, since their normal life-cycle includes both single- and double-stranded stages, their filamentous form allows insertion of variable amounts of foreign DNA, and they are easily manipulated. However, a variety of vectors has been developed that combine the advantages of small, high copy number plasmids with the single-strand generating and filamentous features of the M13 phage.

The Muta-Gene phagemid kit uses the phagemid vectors pTZ18U and pTZ19U.[4] These are plasmids derived from pUC18 and pUC19[5] into which the single-stranded replication origin of the phage f1 has been incorporated. They also contain the multiple cloning site (or polylinker), the promoter for T7 RNA polymerase, and the segment of the *lac*Z gene which permits colormetric screening for inserts. The phagemid replicates as an ordinary double-stranded circular plasmid until its host is superinfected with a helper phage derived from f1 or its close relative M13. The proteins encoded by the superinfecting phage act at the single strand origin and cause the phagemid DNA to be replicated, packaged, and extruded from the cell as if it were a single-stranded phage. Infectious filamentous particles are extruded from the cells containing single-stranded phagemid DNA. The Muta-Gene phagemid kit also supplies a specially designed helper phage, M13KO7,[6] which grows well on its own, but in the presence of a phagemid does not replicate as efficiently as the phagemid. The products of the infection, therefore, contain a relatively large proportion of phagemid and small proportion of helper phage.

The approach used in the Muta-Gene phagemid kit, outlined in Figure 1, is to clone the DNA to be mutated into the phagemid pTZ18U (or pTZ19U), and introduce this recombinant plasmid into the *E. coli dut, ung* strain CJ236. Phagemid particles, whose single-stranded DNA contains some uracils, are produced from these cells by superinfection with the helper phage M13KO7. The uracil-containing DNA is purified, used as template in an *in vitro* mutagenesis reaction, and transformed into a strain with a functional uracil N-glycosylase, thus selecting against the parental strand. If this recipient cell carries an F' plasmid, superinfection with M13KO7 will generate single-stranded particles for sequence analysis.

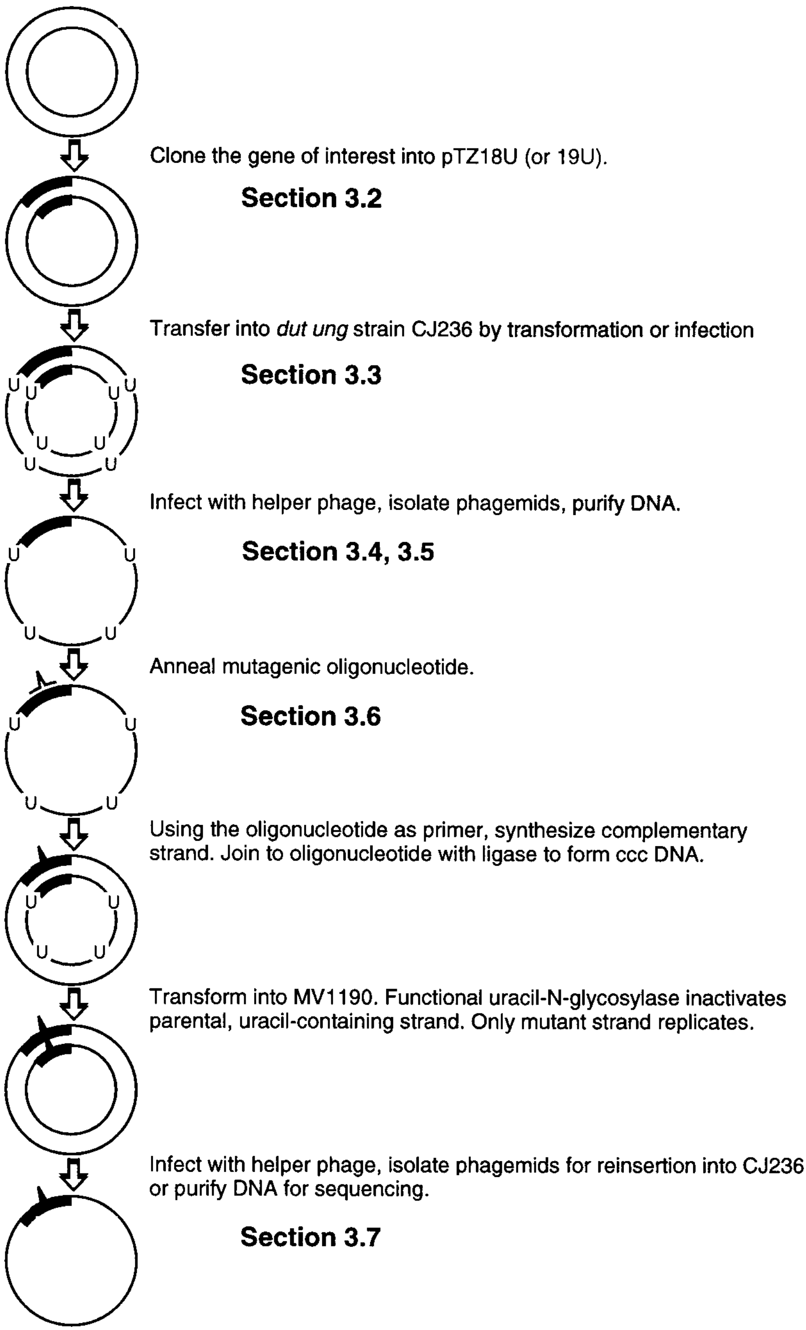


Fig. 1. Steps involved in *in vitro* mutagenesis using the Muta-Gene phagemid kit.

Table 1. Kit Components

1. **pTZ18U and pTZ19U plasmid DNAs:** 5 µg of each at a concentration of 0.1 µg/µl in TE (10 mM Tris-HCl, pH 8, 1 mM EDTA).
2. **M13KO7 helper phage:** 1 ml of a stock at 1–5x10^{11} pfu/ml.
3. ***E. coli* CJ236:** *dut, ung, thi, relA;* pCJ105 (Cm^r), glycerol stock.
4. ***E. coli* MV1190:** Δ(*lac-pro*AB), *thi, sup*E, Δ(*srl-rec*A)306::Tn10 (tet^r) [F: *tra*D36, *pro*AB, *lac* $I^qZ\Delta M15$], glycerol stock.
5. **10x Annealing buffer:** 200 mM Tris-HCl, pH 7.4, 20 mM $MgCl_2$, 500 mM NaCl, 50 µl.
6. **10x Synthesis buffer:** 5 mM each deoxynucleoside triphosphate, 10 mM ATP, 100 mM Tris-HCl, pH 7.4, 50 mM $MgCl_2$, 20 mM dithiothreitol, 50 µl.
7. **T7 DNA Polymerase:** 15 units at 1 u/µl in 20 mM potassium phosphate, pH 7.4, 1.0 mM dithiothreitol, 0.1 mM EDTA, 50% glycerol.
8. **T7 DNA Polymerase dilution buffer:** 20 mM potassium phosphate, pH 7.4, 1.0 mM dithiothreitol, 0.1 mM EDTA, 50% glycerol, 100 µl.
9. **T4 DNA Ligase:** 75 Weiss units in 10 mM Tris-HCl, pH 7.5, 50 mM KCl, 0.1 mM EDTA, 1 mM dithiothreitol, 0.02% BSA, 50% glycerol, 25 µl.
10. **Amber phagemid:** 50 µl of a phage stock of a pTZ18U derivative with an amber (stop) mutation in the *lacZ'* segment which results in a white phenotype in the presence of the chromogenic substrate X-Gal. This stock is provided as a positive control to test the *dut ung* phenotype of CJ236, as described in Section 3.4 and 3.8.
11. **U-phagemid DNA:** 10 µl of 0.2 µg/µl uracil-substituted single-stranded DNA purified from the amber phagemid. This DNA is provided as a positive control for the mutagenesis reaction.
12. **Reverting primer:** 10 µl of 6 pmoles/µl sixteen-base oligonucleotide which reverts the amber mutation (stop) in the U-phagemid DNA to wild-type (Trp). Thus the U-phagemid DNA and the reverting primer can be used as positive controls for the annealing and polymerization reaction as described in Sections 3.6 and 3.8.
13. Specifications sheet
14. Instruction manual
15. Instructions for use of control reagents.

Section 2
Kit Components

All of the components contained in the Muta-Gene phagemid kit are listed in Table 1. The kit contains sufficient reagents to perform 25 oligonucleotide-directed *in vitro* mutagenesis reactions and enough pTZ18U and pTZ19U DNA for about 20 cloning reactions. A detailed description of these components is presented in this section.

2.1 Storage

Upon arrival, store the kit at either -20 ° or -70 °C. The amber phagemid, U-phagemid DNA, and plasmid DNAs should not be thawed and refrozen repeatedly; once thawed, they should be kept at 4 °C. M13KO7 helper phage can withstand repeated freezing and thawing, and should be stored at -20 °C. Immediately before use, spin the tubes for a few seconds in a microcentrifuge to bring the contents to the bottom of the tube. The T7 DNA Polymerase and T4 DNA Ligase should be stored at -20 °C where they are stable for at least 6 months and should not be thawed and refrozen repeatedly.

2.2 pTZ18U and pTZ19U Phagemids DNAs

pTZ18U and pTZ19U[4] are derivatives of the pUC plasmids constructed in the Messing laboratory.[5] In addition to the *col*E1 origin of replication and the gene encoding ampicillin resistance, they carry the *lac* operator and promoter, and the first 145 amino acids of the ß-galactosidase gene (called the *lac*Z' fragment). This protein fragment can complement a truncated host ß-galactosidase fragment (ZΔM15 deletion), producing a functional enzyme. Colonies of the appropriate *E. coli* host carrying the plasmid will be blue in the presence of the chromogenic substrate X-gal. In addition, a multiple cloning site (polylinker) which carries numerous unique restriction enzyme recognition sites was incorporated near the beginning of the *lac* DNA. This polylinker contains unique restriction sites for *Eco*R I, *Sst* I, *Kpn* I, *Xma* I, *Sma* I, *Bam*HI, *Xba* I, *Sal* I, *Acc* I, *Hinc* II, *Pst* I, *Sph* I, and *Hind* III (Figure 2). Insertion of DNA into these sites disrupts the DNA encoding the ß-galactosidase fragment and results in colorless (or pale blue) colonies when plated in the presence of IPTG and X-gal. The insertion of the f1 origin of replication permits, by superinfection with a helper phage, the formation of phage-like particles containing single-stranded DNA. The plus strand of pTZ18U and pTZ19U is packaged. A portion of the sequence of the plus strand is shown in Figure 2. Because these vectors contain features of both plasmids and phage, they are referred to as phagemids. In addition, a strong promoter for T7 RNA polymerase has been inserted into the *lac* region without interrupting the coding sequence. This promoter allows the plasmid DNA to be used to generate high levels of insert-specific RNA by *in vitro* transcription. Other phagemid vectors have been used to generate mutations with the Muta-Gene kit, such as pBS± and pGEM-3Zf±. The pTZ18U and pTZ19U plasmids are 2860 bp in size.

Complete sequence information for pTZ18U and pTZ19U can be obtained from GENBANK with accession numbers L37352 and L37382, respectively.

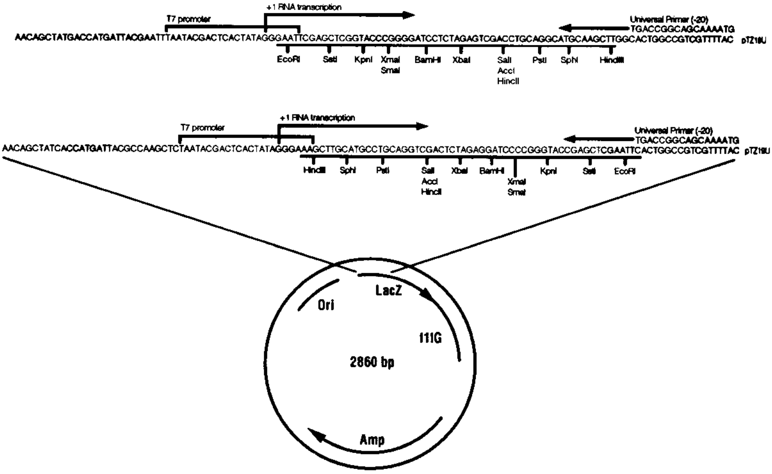


Fig. 2. Diagrammatic representation of the features of the pTZ18U and pTZ19U plasmids. The nucleotide sequence of the multiple cloning region of the plasmids is shown and the restriction sites present in the region noted. Features of the plasmid include the ß-lactamase gene which confers ampicillin (amp) resistance to *E. coli* containing the plasmid, the *lacZ'* fragment of the ß-galactosidase gene, the T7 RNA polymerase promoter region, and the f1 IG region. The last feature makes it possible to generate single-stranded DNA from pTZ plasmids by superinfection of plasmid-containing strains with appropriate helper phages. As shown in the figure, single-stranded DNA generated from pTZ18U or pTZ19U may be sequenced by the dideoxy method utilizing Universal Primer (-20).

2.3 M13KO7 Helper Phage

When wild-type M13 superinfects a host carrying a phagemid, neither are replicated to any extent. This phenomenon is called interference. M13KO7[6] has been constructed from an M13 mutant which partially bypasses this interference. Its own DNA replication has been partially disabled by insertion of DNA carrying the kanamycin resistance gene from Tn903 and the origin of replication of p15A. This, plus the high copy number of the phagemid DNA, leads to packaging of the phagemid at the expense of the helper phage. The kanamycin selects for cells infected with helper.[6] The M13KO7 double-stranded replicative form is 8.7 kb in size.

2.4 Bacterial Strains *E. coli* CJ236 and MV1190

E. coli CJ236 carries the following markers: *dut*-1, *ung*-1, *thi*-1, *rel*Al; pCJ105 (Cm^r). The *dut* and *ung* phenotypes are non-reverting and result in occasional substitution of uracil for thymine in all DNA synthesized in the bacterium.[3] The F' episome pCJ105[7] carries the information for pili construction. Pili are necessary for both phagemid and helper phage attachment to and entry into the bacterial cell. Since F' episomes tend to be lost if not under selective pressure, this strain should be grown in the presence of chloramphenicol.

E. coli MV1190 is Δ(*lac-pro*AB), *thi*, *sup*E, Δ(*srl-rec*A)306::Tn10(tet^r) [F':*tra*D36, *pro*AB, *lac* $I^qZ\Delta M15$]. In this strain, sequences encoding lactose utilization and proline biosynthesis have been deleted from the chromosome. The F' episome carries the proline synthesis genes and a truncated ß-galactosidase gene which produces a protein fragment that can be complemented by the fragment encoded by the pTZ phagemid. Therefore, cells carrying phagemids with an intact fragment sequence will produce blue colonies in the presence of the inducer IPTG and the indicator dye X-gal. Cells carrying phagemids with DNA inserted into the fragment sequence will not produce a functional fragment (or produce a poorly functional fragment) and, hence, will generate colorless colonies. The presence of the proline synthesis genes on the F' allows a selection for the presence of the episome.[8]

2.5 T7 DNA Polymerase

Native T7 DNA polymerase consists of two subunits—an 84,000 dalton polypeptide derived from gene 5 of T7 bacteriophage and a 12,000 dalton polypeptide, thioredoxin encoded by *E. coli*.[9,10] Native T7 DNA polymerase has 5' to 3' polymerase and 3' to 5' exonuclease activity and has been shown to be highly effective in *in vitro* mutagenesis reactions.[11,12] It is a highly processive enzyme and completely copies the uracil-containing template. Its high rate of polymerization allows mutagenic reactions to be performed in less time than with T4 DNA polymerase. Mutagenic efficiencies are comparable with both enzymes. T7 polymerase exhibits a fidelity as good or better than that of T4 polymerase.[25]

T7 DNA polymerase can also alleviate the need for T4 gene 32 protein. When copying templates with secondary structure, T4 DNA polymerase requires gene 32 protein to completely copy the template.

Like T4 DNA polymerase, T7 DNA polymerase does not perform strand displacement[13] and, therefore, will not readily remove the hybridized mutagenic primer. The Klenow fragment of *E. coli* DNA polymerase I, an enzyme previously used in *in vitro* mutagenesis, on the other hand, does exhibit some strand displacement activity. We have found a substantial decrease in the mutation efficiency when using the Klenow polymerase.[14] The difference in the mutation efficiency may be due to the ability of the Klenow polymerase to peel off the mutagenic primer in a strand displacement reaction[15] and hence, copy the template strand eliminating the desired mutation.

We have found that modified T7 DNA polymerase (Sequenase®), which has been chemically treated to inactivate the 3' to 5' exonuclease activity, is not effective in *in vitro* mutagenesis reactions (data not shown). It is also noteworthy that other sources of unmodified T7 DNA Polymerase also do not work in these reactions (data not shown), which indicates that the enzyme may be inadvertently modified during the purification procedure.

2.6 T4 DNA Ligase

T4 DNA ligase is used to ligate the newly synthesized DNA strand to the 5' end of the oligonucleotide primer. The importance of this step has been demonstrated.[12] Using the control reagents supplied with the kit, in the absence of ligase we observed no covalently closed circular DNA on ethidium bromide-stained agarose gels (see Section 3.5), low biological activity, and a low frequency of mutation. Although in some experiments it is possible to get good mutagenesis without *in vitro* ligation,[3,16] we recommend its use since, in at least the experiments described in reference 14, it is necessary.

2.7 Annealing Buffer and Synthesis Buffer

These buffers contain all reagents necessary for annealing the mutagenic oligonucleotide and for synthesizing the complementary strand from the U-containing template. The buffers are functionally tested in oligonucleotide-directed mutagenesis assays.

10x annealing buffer is 200 mM Tris-HCl, pH 7.4 (at 37 °C), 20 mM $MgCl_2$, 500 mM NaCl.

10x synthesis buffer is 5 mM each dATP, dCTP, dGTP, and dTTP; 10 mM ATP, 100 mM Tris-HCl, pH 7.4 (at 37 °C), 50 $MgCl_2$, 20 mM DTT.

2.8 Control Reagents

Several reagents are included in the kit that allow functional testing of the various procedures used with the Muta-Gene phagemid kit. Detailed protocols are given in Section 3.8 and the accompanying specifications sheet.

Amber Phagemid

pTZ18U was altered by oligonucleotide mutagenesis to contain an amber mutation at codon 16 of the *lac*Z' fragment. The UGG codon (tryptophan) was converted to the chain terminating codon UAG. Since the phagemid contains an amber mutation in the *lac*Z' fragment and is, therefore, unable to complement *lac*Z host strains, it yields white colonies on plates containing ampicillin, IPTG, and X-gal (the amber mutation is not significantly suppressed by the *sup*E in MV1190 to give a blue colony). This phagemid stock was prepared on *E. coli* MV1190 and, therefore, does not contain uracils in the packaged DNA. This phagemid allows testing of the functioning of the *dut ung* double mutant CJ236 strain. The amber phagemids are used to infect CJ236. An ampicillin-resistant colony is isolated, grown, and superinfected with helper phage. If the strain is functioning properly, fewer than 1×10^{-4} of the phagemids recovered from the superinfection should form amp-resistant colonies of MV1190 which contains wild-type uracil-N-glycosylase and dUTPase activities. The DNA of this stock can then be purified and used as the template in a mutagenesis experiment with the reverting primer (see below) as a test of the DNA purification procedure.

U-phagemid DNA

This DNA was purified from the pTZ amber phagemid after growth on CJ236 and, therefore, contains uracils. This DNA is included to serve as the control template in *in vitro* mutagenesis DNA synthesis reactions, in conjunction with the reverting primer, as well as for assay of mutagenesis efficiency by transformation into MV1190.

Reverting Primer

A phosphorylated 16-base oligonucleotide which will revert the mutation in the amber phagemid included in the kit. This oligonucleotide has the sequence 5' pGGTTTTCC$\underline{C}$AGTCACG 3' and will revert the amber mutation UAG back to the wild-type UGG codon for tryptophan. Thus, blue colonies are produced on plates containing IPTG and X-gal. The underlined base is the one which is reverted in the amber phagemid by this primer. This oligonucleotide is included in the kit for use as the control primer for the *in vitro* mutagenesis reaction on amber phagemid DNA.

2.9 Additional Reagents for Mutagenesis

Product	Catalog Number
Muta-Gene T7 enzyme refill pack, 25 reactions	170-3582
Unmodified T7 DNA Polymerase, 15 units	170-3583
Muta-Gene phagemid control reagents	170-3577
M13KO7 helper phage, 1.0 ml	170-3578
T4 DNA Ligase, 100 units	170-3111
pTZ18U, 5 µg	170-3560
pTZ19U, 5 µg	170-3561
Electro-competent® *E. coli* strain CJ236	170-3114
Electro-competent *E. coli* strain MV1190	170-3115
Prep-A-Gene® master kit	732-6009
X-Gal, 250 mg	170-3455
IPTG, 1 g	170-3201
C/P Lift® membrane, 85 mm, 50	162-0162
Nitrocellulose membranes, 82 mm, 50	170-3202

Section 3
Protocols

This section describes various procedures used when performing *in vitro* mutagenesis experiments using the Muta-Gene phagemid kit. When using new kit reagents, or if there is any question about any of the reagents that you have prepared, a control reaction using the control reagents included in the Muta-Gene phagemid kit should be performed (see Section 3.8).

3.1 Bacteriology

For detailed instructions on preparation of media, sterile techniques, titering of phage, etc., two manuals from Cold Spring Harbor Laboratory (Cold Spring Harbor, New York, 11724) are very useful: Miller, J. H., (1972) Experiments in Molecular Genetics; and Davis, R.W., Botstein, D. and Roth, J. R. (1980) Advanced Bacterial Genetics.

Media

LB (L broth)

10 g Bactotryptone

5 g yeast extract

10 g NaCl

deionized H_2O to 1 liter

2xYT

16 g Bactotryptone

10 g yeast extract

5 g NaCl

deionized H_2O to 1 liter

Glucose-minimal medium

6 g Na_2HPO_4

3 g KH_2PO_4

0.5 g NaCl

1 g NH_4Cl

deionized H_2O to 1 liter

After autoclaving, add the following filter sterilized solutions

1 ml of 1.0 M $MgSO_4$

0.5 ml of 2% thiamine HCl in d.i. H_2O

10 ml of 20% glucose

H Medium

10 g Bactotryptone

8 g NaCl

deionized H_2O to 1 liter

Agar plates/Top Agar

To prepare agar plates, add 15 g Bacto-Agar to the mixture (1 liter) before autoclaving, cool to 60 °C before pouring. To prepare top agar, add 0.7 g Bacto-Agar/100 ml and autoclave.

Antibiotics/X-gal

Antibiotics, IPTG, and X-gal should be added from stock solutions just before pouring or after liquid medium has cooled below 60 °C.

Ampicillin: The stock solution is 25 mg/ml of the sodium salt in water. Sterilize by filtration and store in aliquots at -20 °C. The working concentration is 50 µg/ml.

Chloramphenicol: The stock solution is 30 mg/ml in 100% ethanol. Store in aliquots at -20 °C. The working concentration is 30 µg/ml for solid media and 15 µg/ml for liquid media.

Kanamycin: The stock solution is 50 mg/ml in water. Sterilize by filtration and store in aliquots at -20 °C. The working concentration is 70 µg/ml.

IPTG (isopropyl-ß-D-thiogalactopyranoside): The stock solution is 100 mM in DI water. Filter sterilize. Add 1 ml/liter.

X-gal (5-bromo-4-chloro-3-indolyl-ß-D-galactoside): The stock solution is 2% in dimethylformamide. Add 2.5 ml/liter.

Growth of Bacterial Strains

Both the phagemids and the helper phage require pili for entry into the cell. These pili are part of the fertility function of *E. coli.* **Since pili are not formed below 35 °C, all growth should take place at 37 °C.** Both the cloning/selecting strain MV1190 and the uracil-substituting strain CJ236 contain plasmids, called F' episomes, which encode the functions required to manufacture pili. Since the cell must exert energy to maintain these episomes, they will be lost if selective pressure is not maintained.

E. coli MV1190 has lost some of the chromosomal genes for synthesis of proline and the entire *lac* operon. The F' carries the *lac* operon except for a small segment of the ß-galactosidase gene **and** the missing genes for proline synthesis. By requiring synthesis of proline, selection is maintained for the F'. Thus, MV1190 should be streaked on a glucose-minimal medium plate, and well isolated colonies on this plate used to inoculate liquid cultures. The defective ß-galactosidase gene can be complemented by the *lac* Z' segment encoded by the pTZ phagemids, thus allowing blue-white screening for inserts.

The F' in *E. coli* CJ236 contains the genes encoding chloramphenicol resistance. Therefore, to maintain the F', CJ236 should always be grown in the presence of chloramphenicol. It should be streaked on an LB (or H) plate containing chloramphenicol, and colonies from this culture used to inoculate liquid medium containing chloramphenicol. A working culture is obtained by streaking from the glycerol stock supplied onto an LB (or H) plate containing chloramphenicol. Well-isolated single colonies from this plate are used to inoculate **every** overnight culture. These plates can be used for 1 month, then a new plate is made by streaking from the original stock supplies, **not** from the plate. The concentration of chloramphenicol **should not exceed** the recommended 15 µg/ml in liquid media. CJ236 has a doubling time about twice that of MV1190.

Both cultures are supplied in 15% glycerol and should be stored frozen (-70 °C).

Helper Phage M13KO7/Titering Phage Stocks

The phage is supplied at a concentration of 1–5x10^{11} pfu (plaque forming units)/ml. The actual titer at the time of bottling is listed on the tube. The phage particles are very stable if kept at 4 °C or below (less than 5% drop in titer over a 3 month period), and can be frozen and thawed repeatedly. If, however, it is desired to retiter the stock, the following procedure is recommended.

1. Prepare a 20 ml culture of MV1190 in LB by inoculating the medium with a well-isolated colony from a glucose-minimal medium plate, and incubating at 37 °C with shaking overnight.
2. Prepare four tubes of top agar by melting it in boiling water or a microwave oven, pipetting 2.5–3 ml agar into sterile tubes, and keeping these tubes at 50°–55 °C until use.
3. Prepare four serial 100-fold dilutions of the M13KO7 stock (10^2-fold, 10^4-fold, 10^6-fold, 10^8-fold) in sterile tubes in LB.
4. Place 0.2 ml of the overnight culture of MV1190 into each of 4 sterile tubes, and add 100 µl each of the last three dilutions into one of these (the last will serve as a control).

5. After 5 minutes at room temperature, pour a tube of top agar into one of the tubes containing culture and phage, mix thoroughly (vortex), and pour onto a plate (fresh plates should always be used), swirling to cover the entire plate with top agar. Repeat with the other tubes.

6. Allow the top agar to harden for 15 minutes, then invert the plates and incubate at 37 °C overnight.

7. The following morning, count the plaques on the plate which has a countable number on it. Since the phages do not kill the cells extruding them, only slow their growth, the plaques will be turbid circles on a denser background.

   the titer = (# of plaques x 10 x dilution factor) pfu/ml.

3.2 Cloning into pTZ Vectors

The first step in the oligonucleotide-directed mutagenesis process is to clone the DNA to be mutated into the pTZ phagemid vector. DNA molecules may be cloned into pTZ vectors by using standard protocols.[17] A restriction map of pTZ indicating the various cloning sites is given in Figure 2.

The cloning reaction should then be transformed into MV1190; and the transformants plated onto H plates containing ampicillin, IPTG, and X-gal. All colonies which grow will contain phagemids which confer ampicillin resistance. The pTZ phagemids have been constructed to maintain the coding frame of the *lac* Z' fragment, so intact phagemids will give rise to blue colonies. Phagemids which have DNA cloned into the polylinker region will produce a defective ß-galactosidase segment or, more likely, none at all, and therefore will generate white or faintly blue colonies.

Two protocols are given for the transformation of cells with DNA. Both rely on cell wall disruption caused by either a high voltage pulse or high concentrations of calcium ions. The procedures described here will yield cells which can be transformed at an efficiency of 10^6–10^9 colonies/µg DNA, sufficient for subcloning into the phagemids and isolating mutants from the *in vitro* mutagenesis reactions.

Method 1: Electroporation

Frozen electro-competent CJ236 and MV1190 cells (catalog numbers 170-3114 and 170-3115) are available for use with Gene Pulser® and E. coli Pulser electroporation systems for high-efficiency transformation. Alternatively, electro-competent cells can be prepared using the strains provided in this kit. Detailed procedures for the preparation of electro-competent cells as well as electroporation parameters can be found in the *E. coli* Pulser instruction manual.

Method 2: $CaCl_2$ Competent Cells

1. Streak MV1190 on a glucose-minimal medium agar plate. Grow at 37 °C for 1 to 2 days until well-defined colonies appear. This plate can be stored in the refrigerator and used to inoculate liquid cultures for 1 or 2 months. The night before you plan to do the transformations and platings, inoculate 20 ml of LB medium with a well-isolated colony. Grow overnight at 37 °C with shaking.

2. Read the $O.D._{600}$ of the overnight culture. A 1/10 dilution gives an accurate reading.

3. To prepare competent cells, inoculate 40 ml of LB medium with enough of the overnight culture to give an initial absorbance reading of approximately 0.1. The rest of the overnight culture can be saved on ice or in a refrigerator for use as a plating culture for titering the helper phage. Incubate with shaking at 37 °C.

4. Inoculate 200–250 ml of LB in a 500 ml Erlenmeyer flask to an $O.D._{600}$ of 0.1 from an overnight culture of MV1190, prepared as described in the previous procedure. Incubate with shaking at 37 °C.

5. When the $O.D._{600}$ reaches 0.9, harvest the culture by centrifugation at 4 °C. For easier resuspension, centrifuge only long enough to pellet the cells (5 minutes at 5,000 rpm in a Sorvall Superspeed centrifuge). Carefully pour off the supernatant and drain well.

   Resuspend the cells very gently in 50 ml of ice-cold 100 mM $MgCl_2$. If it is difficult to obtain complete suspension, the cells may be gently pipetted up and down with a chilled pipet.

6. Again harvest the cells and drain the pellets well. Resuspend the cells **gently** in 10 ml of cold 100 mM $CaCl_2$ until a smooth suspension is obtained. Add 100 ml of cold 100 mM $CaCl_2$, mix, and keep the cells on ice 30–90 minutes.

7. Again harvest the cells and drain the pellets well. Resuspend in 12.5 ml of cold 85 mM $CaCl_2$ and 15% glycerol.

8. Immediately aliquot the cell suspension in 0.6 ml portions and freeze in dry ice/ethanol. These cells will retain competence for at least 6–9 months if kept at -70 °C. To use, simply thaw **on ice** and proceed as below.

Transformation of $CaCl_2$ Competent Cells

Transformations are performed at two stages of the *in vitro* mutagenesis procedure: **i**) when initially cloning the DNA fragment to be mutagenized into the pTZ phagemids; and **ii**) after synthesis of the mutagenized strand on the uracil-containing template. The transformation protocol described here may be used in either case. Slight differences for the two situations are noted in the instructions. This procedure is not applicable to electro-competent cells.

1. For each transformation, place 0.3 ml of competent cells in a cold 1.5 ml sterile polypropylene tube. Keep the tube on ice. If frozen competent cells are used, thaw them **on ice.** Do not allow the temperature of the cell suspension to rise above that of ice, since competence will be rapidly lost.
2. To the competent cells, add 1–10 ng of DNA from a ligation reaction when cloning into pTZ, or 3–10 µl of an *in vitro* mutagenesis reaction after dilution with TE stop buffer (10 mM Tris, pH 8.0, 10 mM EDTA) as described in Section 3.6. Mix gently, and hold on ice for 30–90 minutes. DNA uptake appears to increase slowly from 30 to 90 min. No further increase is observed after that time.
3 Heat shock the cells by placing the tubes in a 42 °C water bath for 3 minutes, then return the cells to ice for at least 5 minutes.
4. The cells must now be grown for a period to allow expression of ampicillin resistance. To each transformation mix, add 1 ml of LB and incubate with agitation for 1 hour at 37 °C. Then spread 10–100 µl onto H plates containing ampicillin, IPTG, and X-gal (see Section 3.1) and incubate at 37 °C overnight.

The following morning, colonies should be visible. It is important not to leave the plates in the incubator longer than overnight. Small satellite colonies of untransformed cells arise near ß-lactamase-secreting colonies and, if allowed to grow, can obscure true transformants. In any case, the isolates should be colony-purified by replating on ampicillin at least through one cycle.

If the transformants are from cloning reactions, there will be blue colonies which carry intact pTZ phagemid, and white colonies which carry pTZ with a DNA insert. Pick white colonies, re-isolate, and verify presence of the insert by standard methods.[18]

All colonies from the *in vitro* mutagenesis reactions will be white (or pale blue). After re-isolation, the colonies are grown up and screened, usually by sequencing reactions, for the presence of the intended mutation.

3.3 Transformation or Infection of Insert-Carrying Phagemid into CJ236

Since the basis of the selection against the non-mutated strand in this procedure is the inviability of uracil-containing genomes in cells with an active uracil N-glycosylase, it is necessary to first prepare DNA containing uracil. This is done in the *dut* (dUTPase), *ung* (uracil N-glycosylase) strain CJ236. Because of the defective enzymes in this strain, **all** DNA synthesized in it contains some uracil.

First, the phagemid must be introduced into CJ236. The phagemid can be introduced into CJ236 either by transformation or infection. In either case, cells containing the phagemid are first selected on plates containing ampicillin, and then restreaked and maintained on plates containing both ampicillin and chloramphenicol. For best results, CJ236 colonies should not be stored on a plate for longer than one week. For longer storage the cells should either be restreaked, retransformed or stored as a glycerol stock.

Once an isolate of CJ236 is obtained which contains the phagemid, single-stranded uracil-containing phagemid DNA is generated by superinfecting the culture with M13KO7 helper phage.

Method 1: Transformation

1. Grow CJ236 in the presence of chloramphenicol as described in Section 3.1, and prepare competent cells by the procedure described in Section 3.2. Alternatively, frozen electro-competent CJ236 and MV1190 cells are available for use with Gene Pulser and E. coli Pulser electroporation systems for high-efficiency transformation.
2. Prepare double-stranded phagemid DNA from an isolate containing the DNA to be mutated. Bio-Rad's Quantum Prep Mini-Prep kit is ideal for this. DNA isolated from 1–5 ml culture is more than sufficient.
3. Transform CJ236 with the phagemid DNA as described in Section 3.2. Plate the transformants onto either H plates or LB plates containing ampicillin (see Section 3.1). Chloramphenicol can be added to the plates optionally. However simultaneous selection with ampicillin and chloramphenicol at this stage can sometimes result in zero transformants. Note that because CJ236 has a functional *lac* operon, **all** colonies will be blue in the presence of IPTG and X-gal, so it is not useful to include these indicators in the plates. Incubate the plates overnight at 37 °C. The colonies obtained from this transformation are used to prepare uracil-containing phagemid (Section 3.4).
4. Restreak a colony on a plate containing ampicillin and chloramphenicol.

Method 2: Infection

Infectious phagemid particles are produced when cells carrying the phagemid are superinfected with helper phage. The helper phage produces proteins which act on the f1 replication origin of the phagemid and cause single-stranded circular DNA to be synthesized and extruded from the cell encased in helper-phage encoded proteins. These phage-like particles can infect cells carrying pili in exactly the same manner as the phage. Upon entry into the infected cell, a second DNA strand is made and the phagemid resumes its plasmid-like replication.

1. Grow a small culture (5 ml) of the transformed isolate in LB + ampicillin until the $O.D._{600}$ is between 0.1 and 0.4 (early log phase).

2. Make a 1/100 dilution of the M13KO7 helper phage stock in TE. Add 5 µl of this dilution to the culture and continue incubation at 37 °C for 1 hour.

3. Spin 1 ml of the culture in a microcentrifuge for 2–3 minutes to remove bacteria and save the supernatant which contains the infectious phage particle.

4. Meanwhile, grow a 20 ml culture of CJ236 in LB + chloramphenicol at 37 °C with aeration to an $O.D._{600}$ of 0.3–0.35. Inoculate this with 10 µl of the supernatant and continue incubation for 2 hours.

5. Dilute the culture 1/10, 1/100, and 1/1,000 in LB and spread 50 µl of each dilution on H or LB plates containing ampicillin. Incubate the plates at 37 °C overnight. Do not incubate longer because of the growth of satellite colonies.

6. From the plate with the least number of colonies, pick one and restreak on a plate containing ampicillin and chloramphenicol.

3.4 Growth of Uracil-Containing Phagemids

One of the major factors in performing successful mutagenesis experiments using this system is the preparation of uracil-containing phagemids carrying the fragment to be mutagenized. We have determined through rigorous testing that the strain CJ236 included in the Muta-Gene kit is *dut ung*. Controls for checking the phenotype of the strain are described in Section 3.8. In the event that a particular insert does not package efficiently, it may be possible to obtain phagemid particles by reversing the orientation of the insert.[19]

1. Streak out CJ236 containing pTZ with the desired insert (prepared in Section 3.3) onto an LB plate containing chloramphenicol (see Section 3.1). Grow at 37 °C until distinct colonies appear.

2. Pick an isolated colony and place in 20 ml of LB containing 15 µg/ml chloramphenicol. Incubate with shaking at 37 °C overnight.

   NOTE: See question number 1 in the section on Troubleshooting (section 5).

3. Inoculate 50 ml of 2xYT media containing ampicillin and chloramphenicol with 1 ml of the overnight culture of CJ236 containing phagemid. Incubate with shaking at 37 °C. Only 30 ml of this culture will be used to isolate uracil-containing phagemid DNA. The remainder is surplus for reading O.D.

4. Grow to an $O.D._{600}$ of 0.3, which will take from 1 hour to 4 hours. This corresponds to **approximately** 1×10^7 cfu/ml. Add helper phage M13KO7 to obtain an M.O.I. (multiplicity of infection) of around 20, *i.e.,* 20 phage/cell. The original titer of the M13KO7 stock is listed on the tube. It is generally between 1–5 x 10^{11} pfu/ml. If it is desired to retiter the phage, instructions can be found in Section 3.1.

5. Incubate with shaking at 37 °C for 1 hour. then add kanamycin (50 mg/ml stock) to a final concentration of 70 µg/ml. Continue incubation for 5-6 hours. Never allow the culture to be incubated overnight, this results in the presence of lysis products in the final DNA preparation.

6. Transfer 30 ml of the culture to a 50 ml centrifuge tube. Centrifuge at 17,000 x g (12,000 rpm on the Sorvall SS-34 rotor) for 15 minutes. Transfer supernatant which contains the phagemid particles to a fresh centrifuge tube. Recentrifuge this at 17,000 x g for 15 minutes at 4 °C.

7. Transfer the second supernatant to a fresh polyallomer centrifuge tube and add 150 µg RNase A. See reference 18 for details on preparing DNase free RNase. It is also helpful to add RNAse T1 to a final concentration of 20 µg/ml at this stage. Incubate at room temperature for 30 minutes.

8. To the phagemid-containing supernatant, add ¼ volume 3.5 M ammonium acetate/20% PEG-6,000 and mix well. Incubate on ice for 30 minutes.

9. Collect the phagemids by centrifuging at 17,000 x g for 15 minutes. Decant the supernatant and discard. Drain the pellet well. Resuspend the pellet in 200 µl of high salt buffer (300 mM NaCl, 100 mM Tris, pH 8.0, 1 mM EDTA). Transfer the resuspended phagemids to a 1.5 ml microcentrifuge tube and chill on ice for 30 minutes. Centrifuge for 2 minutes to remove insolubles and transfer supernatant to a fresh tube. Store at 4 °C. DNA from this preparation should be purified within 1 week. Reserve 20 µl of this stock to determine phagemid uracil incorporation.

Test For Uracil Incorporation

Since the phagemids contain uracil in their DNA, they should survive far more readily in a bacterium without an active uracil N-glycosylase than in one with an active enzyme. If the phagemids were produced with a significant amount of uracil in their DNA, they should produce more ampicillin-resistant colonies in CJ236 than in MV1190. This procedure is optional but can provide important data for troubleshooting.

1. Grow overnight cultures (ONC) of MV1190 and CJ236. Add 0.5 ml of the MV1190 ONC to 50 ml of 2xYT. Add 1 ml of the CJ236 ONC to another 50 ml of 2xYT. Incubate with shaking at 37 °C until the $O.D._{600}$ is 0.3 to 0.35. If one culture reaches this O.D. before the other, place it on ice.

2. Add 10 µl uracil-containing phagemid stock to each culture. Grow at 37 °C for 2 hours with shaking. Dilute the MV1190 culture 10-fold and 100-fold, and spread 50 µl of each dilution and the undiluted culture onto ampicillin-containing H or LB plates. Dilute the CJ236 culture 10^3-fold, 10^4-fold, and 10^5-fold and spread 50 µl of these dilutions onto ampicillin-containing plates. Incubate at 37 °C overnight.

There should be 10^2 to 10^4-fold fewer colonies from the MV1190 culture than from the CJ236 culture.

3.5 Extraction of Phagemid DNA

After a suitable phagemid stock that has uracil-containing DNA has been obtained, the DNA must be purified by extraction prior to its use as template in the *in vitro* mutagenesis reactions. The procedure for extracting the viral DNA is described in this section. The extractions are conveniently performed in standard 1.5 ml microcentrifuge tubes. Alternatively, mutagenesis-grade templates can be purified without phenol/chloroform extractions with the Prep-A-Gene Master Kit by adding 620 µl of TE buffer to the phage (step 9) and follow the Prep-A-Gene instruction manual, Section 3.2, from step 2.

1. Extract the entire 200 µl phagemid stock 2x with an equal volume of neutralized phenol (see reference 17 for neutralization procedure), 1x with phenol/chloroform (1:1:1/48 phenol: chloroform:isoamyl alcohol), and several times with chloroform/isoamyl alcohol. Continue the chloroform/isoamyl alcohol extractions until there is no visible interface, then once more. It is important to vortex each extraction for 1 minute. The yield of DNA can be increased 30–50% by backextracting each step: add 100 µl of TE (10 mM Tris, pH 8.0, 1 mM EDTA) to the first phenol extraction tube, vortex, add resultant aqueous phase to the next phenol extraction tube, vortex, and so on.

2. Pool the aqueous phases, add ⅒ vol 7.8 M ammonium acetate and 2.5 vol ethanol. Keep at -70 °C at least 30 minutes.

3. Centrifuge 15 minutes in the cold, carefully remove the supernatant, wash the pellet gently with 70% ethanol, and resuspend the pellet in 20 µl of TE. Avoid dissolving any residue that may cling to the side of the tube.

4. Transfer the dissolved DNA to a fresh tube. Run a small aliquot on an agarose gel with a known amount of single-stranded DNA to determine the DNA concentration. Typically, a few µg DNA are obtained. Since only 0.2 µg is used in an *in vitro* mutagenesis reaction and this reaction should yield tens of thousands of transformants, it is not necessary to isolate large amounts of DNA. The quality of the single-stranded phagemid DNA we typically purify with this procedure is shown in Figure 3. This experiment also demonstrates that we usually obtain very little endogenously primed (no added primer) synthesis when this DNA is used in an *in vitro* mutagenesis reaction (see Section 3.6).

3.6 Synthesis of the Mutagenic Strand and Transformation of the Reaction Products

After preparation of single-stranded uracil-containing DNA from the phagemids, *in vitro* mutagenesis reactions may be performed using it as template, and the reaction products transformed into the uracil N-glycosylase-containing strain MV1190. This is accomplished by priming the synthesis of the single-stranded DNA with the oligonucleotide containing the sequence of the desired mutation(s). Various strategies for designing oligo-nucleotides for the purpose of inserting mutations have been described.[20,21]

This section is divided into five parts: i) phosphorylation of the mutagenic oligonucleotide (we found that the frequency of mutagenesis dropped three-fold when the primer was not phosphorylated); ii) annealing of the mutagenic primer to the template; iii) synthesis of the complementary strand; iv) gel analysis of the reaction products; and v) transformation of the reaction products.

Phosphorylation of the Oligonucleotide

The oligonucleotide should be lyophilized after synthesis and resuspended in water at 10–20 pmol/µl The following procedure has been taken from reference 22 and been successfully used at Bio-Rad.

1. Prepare the following reaction in a sterile 500 µl microcentrifuge tube:

Component	Volume	Final Conc. or mass
oligonucleotide	varies	200 pmol
1 M Tris, pH 8.0	3 µl	100 mM
0.2 M $MgCl_2$	1.5 µl	10 mM
0.1 M DTT	1.5 µl	5 mM
1 mM ATP (neutralized)	13 µl	0.4 mM
sterile water	varies	to total volume of 30 µl

2. Mix well.
3. Add 4.5 units of T4 polynucleotide kinase to the reaction mixture.
4. Mix and incubate at 37 °C for 45 minutes.
5. Stop the reaction by heating at 65 °C for 10 minutes.
6. Dilute the oligonucleotide to 6 pmol/µl with TE (10 mM Tris, pH 8, 1 mM EDTA), store frozen.

Annealing of the Mutagenic Oligonucleotide

1. Prepare the following reaction mix in a 500 µl microcentrifuge tube:

Component	Volume	Final Conc. or mass
uracil-containing DNA	varies	200 ng (0.2 pmol)
mutagenic oligonucleotide	varies	6-9 pmol
10x annealing buffer	1 µl	20 mM Tris-HCl (pH 7.4), 2 mM $MgCl_2$, 50 mM NaCl
water	varies	so that the total volume of the reaction is 10 µl

Note: The molar ratio of primer to template in this reaction is between 20:1 and 30:1 for a 16-mer oligonucleotide and a 2,860 base template. These are the sizes of the primer and template included with the control reagents. We have found that high ratios can interfere with subsequent ligation. Also, very high ratios of primer to template can result in a significant level of spurious priming from secondary hybridization of primer to the template.[14]

2. Prepare a second reaction mixture containing all the above ingredients except the primer. This is a control reaction which will test for non-specific endogenous priming caused by contaminating nucleic acids in the template preparation. This test is important because endogenous priming may result in lowered mutagenesis efficiency. Properly prepared templates should result in little, if any, ccc DNA synthesized in the absence of added primer (see Figure 3).
3. Place the reaction mixtures in a 70 °C water bath. Allow the reactions to cool in the water bath at a rate of approximately 1 °C per minute to 30 °C over a 40 minute period. After this, place the reactions in an ice-water bath. **These annealing conditions have been optimized for use with the control reagents supplied with the Muta-Gene phagemid kit. It may be necessary to optimize conditions for the particular oligonucleotide and template used.**

Synthesis of the Complementary DNA Strand

The first time the kit is used, reactions with the control reagents (as described in section 3.8) should be performed in parallel with the reactions listed below. This will serve both to demonstrate that the enzymes are functioning properly and also provide samples that can be electrophoresed on a gel to demonstrate the various DNA forms.

1. The T7 DNA polymerase is supplied at 1.0 unit/µl and must be diluted to 0.5 unit/µl for use. Make a 1 to 1 dilution of the T7 DNA polymerase using the T7 DNA polymerase dilution buffer (provided). Since the enzyme is not as stable at the lower dilution, only dilute an amount necessary for immediate use.
2. With the reactions still in the ice-water bath, add the following components in the order listed:

Component	Volume	Final conc. or mass
10x synthesis buffer	1 µl	0.4 mM each dNTP 0.75 mM ATP 17.5 mM Tris-HCl (pH 7.4) 3.75 mM $MgCl_2$ 1.5 mM DTT
T4 DNA ligase	1 µl	3 units
T7 DNA polymerase (diluted)	1 µl	0.5 units

Note: The reaction conditions for DNA synthesis are (including the contributions from both annealing and synthesis buffers): 23 mM Tris (pH 7.4), 5 mM $MgCl_2$, 35 mM NaCl, 1.5 mM DTT, 0.4 mM dATP, dCTP, dGTP, and dTTP, 0.75 mM ATP, plus the nucleic acids and enzymes, as given above.

3. Incubate the reactions on ice for 5 minutes (to stabilize the primer by initiation of DNA synthesis under conditions that favor binding of the primer to the template[3]), then at 25 °C for 5 minutes, and finally at 37 °C for 30 minutes.
4. After 30 minutes, add 90 µl of TE stop buffer (10 mM Tris, pH 8.0, 10 mM EDTA) to the reaction and stop the reaction by freezing. The reaction is stable at -20 °C for at least 1 month for use in the subsequent transformation.

Gel Analysis of the Reaction Products

The reaction products should be analyzed on a 1% agarose gel in 1x Tris-acetate buffer that contains 0.5 µg/ml ethidium bromide. The ethidium bromide binds to covalently closed circular (ccc), relaxed DNA and causes positive supercoils. This condensation causes the DNA to migrate more rapidly through the gel. The second strand synthesis reaction results in the formation of relaxed ccc DNA, and, hence, a band migrating slower than single-stranded (ss) plasmid DNA indicates successful conversion to the biologically active ccc DNA (see Section 2.5). An example of an ethidium bromide-agarose gel analysis of an *in vitro* mutagenesis experiment is presented in Figure 3. In the absence of primer, little single-stranded template is converted to ccc DNA, indicating that the template is not contaminated with impurities which can prime complementary strand DNA synthesis. At primer-to-template ratios of 20:1 to 30:1 we typically convert 50–80% of the ss DNA template to primarily double-stranded material which includes both ccc and oc (open circle) DNA forms, 10–50% of this material is converted to ccc DNA when ligase joins the newly synthesized strand to the 5' end of the mutagenic oligonucleotide. Production of ccc DNA in the absence of added primer is indicative of contaminated DNA template, and new template should be purified from a fresh phagemid stock. Failure to produce ccc DNA when primer has been added can result from a variety of causes. Common problems include inactive T7 DNA polymerase, failure of the primer to hybridize to the template, and improperly set-up reactions (see reference 3 for a complete discussion of possible causes).

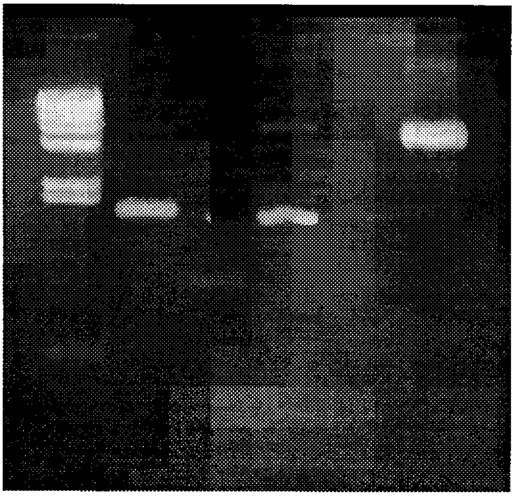
Gel analysis of the reaction products may be performed as follows.

1. Add to 9 µl of each *in vitro* mutagenesis reaction, 1 µl of gel loading buffer (1% SDS, 0.25% bromophenol blue, 50% glycerol). Reactions that should be analyzed on the gel include the no-primer control and the complete reaction.

2. Set up one microcentrifuge tube containing 100 ng of the single-stranded form and another tube containing 100 ng of the double-stranded (ccc) form of the phagemid that is being mutagenized. Bring the volume of each tube to 9 µl with TE and add 1 µl of gel loading buffer. This DNA will serve as the marker to aid in identification of ccc DNA. If the control reactions have also been performed (as described in section 3.8), they should be electrophoresed on the same gel.

3. Run the samples on a 1% agarose gel containing 0.5 µg/ml ethidium bromide. A mini-gel is convenient for this purpose and electrophoresis is carried out until the bromophenol blue dye has migrated about halfway through the gel. It is not necessary to add ethidium bromide to the electrophoresis buffer if the gel is not run any longer than this. If electrophoresis is carried out beyond this point, the front of ethidium in the gel (which travels toward the negative pole) passes the fastest-migrating DNA bands and the ethidium bromide bound to the DNA is stripped off.

4. Destain the gel for 15 minutes in distilled water (there is no need to stain the gel, due to the presence of ethidium bromide in the gel) and then photograph the gel. Interpretation of the pattern has been discussed previously in this section and will be aided by reference to Figure 3. If little or no ccc DNA is observed in the reaction with added primer, refer to Section 3.8 (Use of Muta-Gene Phagemid Control Reagents). If there is a substantial amount of ccc DNA present in the reaction with primer added, compared to the no-primer control, the reaction products may be transformed as described later in this section.



lane	1	2	3	4	5	6
primer	-	-	-	+	-	+
template	-	-	+	+	+	+
polymerase	-	-	-	+	+	+
ligase	-	-	-	+	+	-

Fig 3. Band identification. Reactions with single-stranded U-phagemid DNA template for the purpose of illustrating the various DNA forms that can be seen on a 1.0% Molecular Biology Certified Agarose gel (162-0133).

Lane 1 contains *Hin*dIII-cleaved lambda DNA as a size standard (170-3470).

Lane 2 contains double-stranded phagemid DNA alone (supercoiled). The supercoiled (Form I) and a small amount of open circular (Form II) is seen. Phagemid DNA is approximately 2.7 kb in size.

Lane 3 contains single-stranded U-amber phagemid DNA alone. The primary band runs between the 564 and 2027 bp fragments of the *Hin*dIII-cleaved lambda DNA size standard (the smallest two bands in lane 1). Faint bands comigrating with the bands from lane 2 indicate that a small amount of lysis occurred in this preparation, releasing the replicative forms of the U-amber phagemid DNA that exist in the cell. Additional faint bands represent: **1**) single-stranded M13KO7 helper phage DNA located between the supercoiled and the open circular forms of double-stranded U-amber phagemid DNA; **2**) double-stranded M13KO7 RF (replicative form) located above the open circular form of U-amber phagemid DNA; and **3**) bacterial genomic DNA located near the well. Note that these last three bands are not altered in intensity when treated with polymerase and ligase, as seen in lanes 4, 5, and 6.

Lane 4 is the U-amber phagemid DNA, annealed to the Reverting Primer and incubated with T7 polymerase and T4 DNA ligase. The single-stranded template has disappeared completely and has been converted into a small amount of open circular form, and a large amount of supercoiled form which co-migrates with the supercoiled DNA seen in Lane 3.

Lane 5 is similar to lane 4, but with no added primer (this is the no primer control). Comparing lane 5 to lane 3, one sees a small amount of conversion of the single-stranded form to open circular and supercoiled forms. This conversion is probably the result of sheared genomic DNA acting to prime DNA synthesis, and may reduce mutagenesis efficiency slightly.

Lane 6 is similar to Lane 4, but with no added T4 ligase. Only the open circular form is seen, since the absence of ligase prevents the sealing of the gap between the 5' end of the Reverting Primer and the 3' end of the complementary strand which converts the open circular form to the supercoiled form.

Additional Notes:

A) Very small bands (less than one kb) can either be deletion mutants or RNA. See the Troubleshooting Guide.

B) Predicting the migration of single-stranded DNA templates can be difficult. On this gel, the single-stranded form of U-amber phagemid DNA runs faster than the double-stranded form, to a distance where a 1.0 kb double-stranded DNA fragment would migrate.

Transformation of Reaction Mixture

Once the gel analysis indicates that a successful *in vitro* synthesis has been obtained, the next step is to transform the reaction products into the *E. coli* strain MV1190. This strain has an active uracil N-glycosylase which will inactivate the uracil-containing parental strand, thus enriching for the mutant strand.

1. Prepare competent MV1190 cells as described in Section 3.2.

2. Following the instructions in Section 3.2, transform 10 µl of the reaction mixtures into 0.3 ml competent cells. As controls, transform 3 µl of the no primer reaction and 1 ng of either pTZ18U or 19U or the parental plasmid.

3. Spread 10 and 100 µl of the transformations on LB + ampicillin plates. The inclusion of IPTG and X-gal is not useful here since all the colonies will carry pTZ with an insert and will therefore all be white (or pale blue).

4. After allowing the bacterial suspension to be absorbed into the plates for about 10 minutes, invert them and incubate at 37 °C overnight after which colonies should be visible. We typically obtain 30 to several hundred colonies depending on the batch of competent cells and the efficiency of the *in vitro* synthesis reaction. Our competent cells usually yield around 500 colonies with 50 µl of the final suspension after transformation with 1 ng pTZ DNA. The number of colonies obtained in the no-primer reaction should be less than 20% of that obtained with primer. If results similar to this are obtained, colonies can be picked, purified, and screened for those containing the desired mutant phagemid (Section 3.7).

3.7 Analysis of Transformants by DNA Sequencing

Because of the strong selection against the uracil-containing, non-mutant parental DNA strand, typically more than 50% of the colonies obtained from the transformation of MV1190 carry mutant phagemids. This high efficiency of mutant production allows identification of mutants by direct DNA sequence analysis. This analysis is facilitated by an important property of phagemids, namely their existence in single-stranded form.

To prepare template for sequencing, grow a 50 ml culture from a purified colony of MV1190 containing a putative mutant as described in Section 3.4, except do not include chloramphenicol in the growth medium (chloramphenicol was used to insure the presence of the F' plasmid in CJ236). Infect with M13KO7 and concentrate phagemids as described in that section. Extract the DNA as described in Section 3.5. This template may now be sequenced by standard Sanger dideoxy reactions. Figure 4 is typical of the data we obtain with control reagent templates prepared in this manner.

Note: The pTZ phagemids contain the *lac* Z' region of pUC18 or pUC19 (see Section 2.2, Figure 2). Therefore the Universal Primer (-20) can be used if the mutation is within 250 bases of the 3' end of the insert. Otherwise, an oligonucleotide complementary to a position closer to the mutation may be used as sequencing primer.

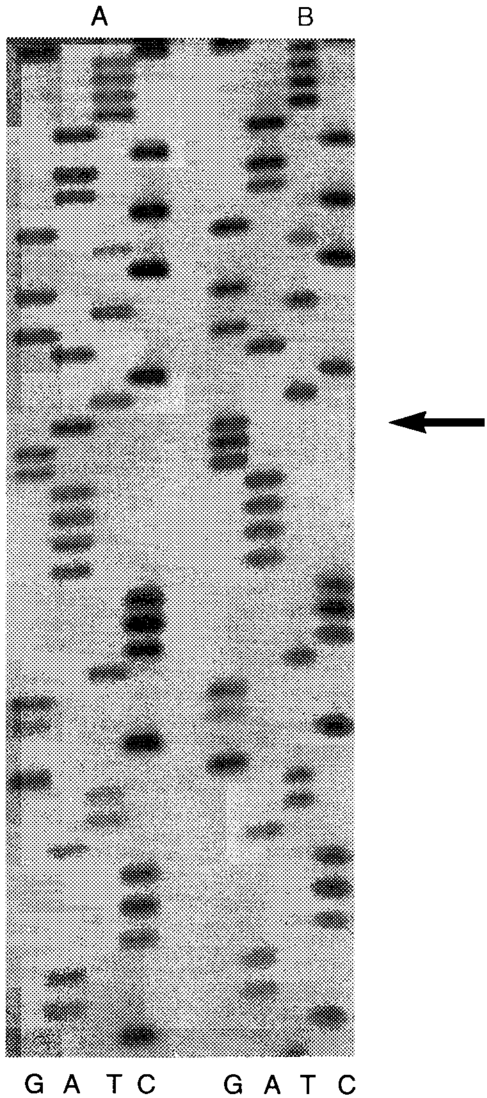


Fig. 4. Analysis of control reagents by DNA sequence analysis. A) The template was 0.5 μg of uracil-substituted pTZ18U DNA (U-phagemid DNA) encoding an amber (stop) mutation at codon 16 of the *lacZ*' fragment, which yields colorless colonies in the presence of X-Gal and IPTG. **B**) The template was 0.5 μg of pTZ18U DNA from a control mutagenesis reaction with the reverting primer (5' pGGTTTTCCCAGTCACG 3') and the U-phagemid DNA. The primer reverts the amber (UAG) mutation at codon 16 back to Trp (UGG), which yields blue colonies in the presence of X-Gal and IPTG. See Section 3.8 for mutagenesis conditions. Templates were sequenced with Bst DNA polymerase and [α-^{35}S]dATP, separated on a 6% polyacrylamide/7M urea gel at 55° C using a 21 x 40 cm Sequi-Gen system until the bromophenol blue tracking dye reached the bottom. The arrow indicates the reversion of A (amber) to G (wild-type).

3.8 Use of the Muta-Gene Phagemid Control Reagents

These reagents are included so that the user can test all components of the *in vitro* mutagenesis procedure and try the protocols before committing valuable experimental material. Table 2 lists the reagents and their properties and uses.

Table 2. Muta-Gene Phagemid Control Reagents

Test Reagent	Description	Purpose
Amber phagemid	pTZ phagemids with an amber mutation in the *lacZ'* region. Colonies are white on IPTG/X-gal/ampicillin plates. Grown on MV1190, so do not contain uracil in their DNA.	Test of the dut, ung phenotype of CJ236, and for the complete *in vitro* mutagenesis procedure—the amber mutation will be reverted.
U-phagemid DNA	Uracil-containing single-stranded DNA of the amber phagemid. Grown on CJ236 and purified.	Test of the *in vitro* mutagenesis synthesis reagents and the presence of active uracil-N-glycosylase in MV1190.
Reverting primer	Phosphorylated 16-base oligonucleotide which will revert the amber mutation in *lacZ'*.	Used in conjunction with U-amber phagemid DNA to test functioning of synthetic reagents.

Amber Phagemid

1. Grow an overnight culture of CJ236 in LB plus chloramphenicol.
2. Inoculate 50 ml 2xYT plus chloramphenicol with 1 ml of the overnight culture of CJ236 and incubate at 37 °C until early log phase (O.D.$_{600}$ of 0.3–0.35)
3. Add 10 µl of amber phagemid and incubate at 37 °C for a further 2 hours.
4. As described in Section 3.3, dilute and spread on ampicillin plates and incubate overnight.
5. Proceed as described in Section 3.4 to prepare a uracil-containing stock of the amber phagemid by growing one of the resulting colonies, infecting with helper M13KO7, and concentrating the phagemids.

6. Titer the amber phagemids on both CJ236 and MV1190 as described in Section 3.4. If the titer is high enough on CJ236 and low enough on MV1190 (indicating sufficient uracil in the phagemid DNA), the phenotype of CJ236 is *dut ung* and the F' plasmid is present.

7. If desired, the DNA can be purified from the remainder of the stock (Section 3.5) and used as template in an *in vitro* mutagenesis reaction (see below).

U-phagemid DNA and Reverting Primer

The Quality Control test of these reagents consists of four reactions: one without primer, the other three with increasing amounts of primer. The specifications sheet which is included in the Muta-Gene phagemid kit gives the number of colonies obtained from each reaction and the percent which were mutant using the batch of reagents contained in the kit. A separate sheet of instructions entitled "Use of Muta-Gene Control Reagents" is included with each kit, containing specific instructions for the included lots of control reagents. **If the instructions on this sheet differs from those given below, the instructions on the sheet should be used instead of those below.**

1. Prepare the following reactions in a 0.5 ml microcentrifuge tube (10 µl total final volume).

Component	Reaction #1	Reaction #2
U-phagemid DNA	1 µl (0.1 pmole)	1 µl
reverting primer	0 µl	2 µl (4 pmole)
10x annealing buffer	1 µl	1 µl
Sterile water	to 10 µl total	to 10 µl total

2. Place the reactions in a 70 °C water bath. Allow the water bath containing the reactions to cool in the water bath to 30 °C over a 40 minute period. After this, place the reactions in an ice-water bath.

3. With the reactions still in the ice-water bath, add the following components in the order listed:

   1 µl 10x synthesis buffer
   1 µl T4 DNA ligase (3 units)
   1 µl T7 DNA polymerase (0.5 unit)

   For the most economical use of the enzyme, make a 1:1 dilution of the 1 unit/µl enzyme solution using the T7 DNA Polymerase dilution buffer (provided).

4. Incubate the reaction mixtures on ice for five minutes and then at 25 °C for five minutes. Finally, incubate at 37 °C for 30 minutes.

5. Add 90 µl of stop buffer as in standard protocol, Section 3.6.
6. Analyze the reactions on an agarose gel as described in Section 3.7. Figure 3 shows an analysis of a typical set of *in vitro* mutagenesis reactions.
7. If the reactions were successful, the reaction products are transformed into competent MV1190. Competent cells are prepared as described in Section 3.2, and the reaction products are transformed as described in Section 3.6, except that the transformed cells are grown on plates containing IPTG (5 mM) and X-gal (40 µg/ml) as well as ampicillin. The colonies containing reverted pTZ amber phagemid will be blue. Those containing phagemids that have not been reverted by the mutagenesis reaction will be white.

We typically obtain between 60 and 80% mutants in this particular reaction, and 100–500 colonies/100 µl of the final (after expression) culture.

Section 4
Common Questions

1. *Do I have to use the vector included in the kit if my insert is already cloned into another vector? If I use another vector, what helper phage should I use?*

   Any phagemid should work. Try M13KO7 as the helper phage. If you have any problems with yield of single-stranded phagemid DNA, use the helper phage recommended by the manufacturer of the vector you have chosen.

2. *Do I have to use the cell lines included in the kit?*

   CJ236 is an essential component of this method, since the *dut ung* genotype of CJ236 results in the production of a uracil-substituted single-stranded template. MV1190 is a good cloning strain, but any *rec*A strain that is wild-type for *dut ung* (which most cloning strains are) and can support M13 infection and growth can be used.

3. *How do I design my primer?*

   By definition, the plus strand is the strand that is packaged in the phagemid. In pTZ18U, the plus strand is the strand that reads 5' to 3' in the same direction as the sequence shown on page 6 (from left to right). In other words, it is the strand which reads 5' to 3' from the *Eco*RI site through the *Hin*dIII site. Determine what the orientation of your insert is in the vector, and which strand of your sequence is in the plus strand. Design an oligonucleotide complimentary to that strand. For single point mutations, an oligonucleotide of about 16 bases is sufficient. For a deletion or insertion, we recommend 10-15 bases on either side of the mutation to "clamp" the oligonucleotide to the template.

4. *What type of mutations are possible?*

   Point mutations are readily obtained. Deletions, even quite large ones, present no significant problems, although mutagenesis efficiency may be reduced. Deletions of 950 bases have been reported[26] and even larger deletions are possible. Insertions however should be limited to 12 bases, as larger insertions may significantly reduce the mutagenesis efficiency. The largest reported successful insertion is 27 bases.[24]

5. *What are the sizes of pTZ18U, pTZ19U and M13KO7?*

   pTZ18U and pTZ19U are 2860 bp. M13KO7 is 8.7 kb.

6. *Can I substitute another enzyme for T7 DNA polymerase?*

   T4 DNA polymerase will work (and this enzyme is provided in our original kit version). However, it is not as processive as T7 DNA polymerase. T7 polymerase is less likely than T4 to prematurely terminate at a region of secondary structure. Modified T7 polymerase, the form used in DNA sequencing reactions, will not work. It is necessary to use a T7 polymerase that is prepared specifically for mutagenesis. Other sources of T7 polymerase may exhibit significant strand displacement activity, which results in removal of the mutagenic oligonucleotide following synthesis of the complementary strand. The result is high transformation efficiencies, with extremely low mutagenesis efficiency (no mutants).

Section 5
Troubleshooting

1. *I am seeing polymerization of my template even in the absence of primer. What is wrong?*

   Spurious priming from contaminating RNA or sheared bacterial genomic DNA. This may also be seen as a very low mutation efficiency, or by a high number of colonies resulting from transformation of MV1190 with the no-primer control reaction. These contaminants are due to lysis of the CJ236 during the single-stranded DNA preparation procedure. Using RNAse T1 in the isolation may help minimize the RNA contaminant. However the best solution is to avoid the lysis in the first place. For best results, use a freshly transformed CJ236 colony and carefully follow our directions concerning incubation times and the optimal O.D. of the cells prior to superinfection with the helper phage. If following these directions still produces a contaminated prep, the chloramphenicol should be deleted from the liquid media steps (including the overnight and the 50 ml culture). Some users have experienced slow growth and lysis in the presence of the combination of ampicillin and chloramphenicol simultaneously. It is of critical importance however that the cells be maintained on fresh chloramphenicol plates, to maintain the presence of the pCJ105 plasmid which carries the F transfer operon.

2. *My single-stranded template preparation has several small contaminating bands. What are these?*

   We sometimes see small RNA fragments that RNAse A does not seem to completely digest. If these small bands are seen, treat the sample with RNAse T1 (20 µg/ml final concentration). If a combination of RNAse A and RNase T1 will not remove small contaminating bands, they may be small deletion mutants of your phagemid. These could result if secondary structure in your insert seriously reduces the packaging efficiency of the phagemid, giving the occasional, spontaneously occurring mutant a selective advantage. If the small bands are the predominant DNA forms, the only solution may be recloning the insert in the other orientation and/or into another vector.

3. *My single-stranded template preparation is contaminated with M13KO7 helper phage. Will this cause a problem?*

   It is normal to see a faint band at about 3-4 kb in a single-stranded DNA prep. Contaminating M13KO7 will not generally cause a problem, even if it is the predominant band, as long as enough of your template is present. If however there is very little of your template in a preparation containing much M13KO7, you may not be able to obtain any mutants. This situation can result from either using a very high multiplicity of infection (MOI) in your DNA preparation procedure, using a CJ236 colony which has been stored on a plate for an extended period of time or because of some structural characteristic of your phagemid that interferes with its replication or packaging. Check your MOI and retransform CJ236 with your phagemid, selecting a new colony. If a preparation from a fresh transformation yields the same results, than the only solution may be recloning the insert into the other orientation and/or into another vector.

Section 6
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